



**Tech
Tra
PlastiCE**

**Strengthening University tech
transfer capabilities to support
circular economy value chains
for plastics in Latin America**

**ECHO 2026 – Sustainability and the
Circular Future with UniGR-CIRKLA**

Interreg



Cofinancé par
l'Union Européenne
Kofinanziert von
der Europäischen Union

Grande Région | Großregion



Assistant Prof. Fabio A. Cruz Sanchez
General Coordinator
Sept 15/2026



**UNIVERSITÉ
DE LORRAINE**

**LORRAINE
INP**

Ensgsi
NANCY



**Co-funded by
the European Union**

Plastic Problem: View from Global South

Paradox:

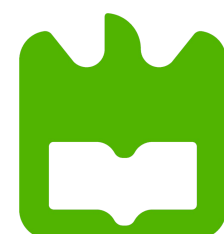
Latin America is a crucial provider (11% of the world's raw materials by weight) → Exponential rise of global material demand is a major driver of the region's environmental degradation and social liabilities

Bottom-up approaches → Historical Grassroot and waste picker associations

Interaction Academia - Industry is not sufficient



Consortium (Erasmus+ CBHE)



Columbus
ASSOCIATION



UNS
UNIVERSIDAD
NACIONAL DEL SUR



Universidad
Nacional
de Córdoba



UNIVERSIDAD
DE SANTIAGO
DE CHILE



PONTIFICIA
UNIVERSIDAD
CATÓLICA DE
VALPARAÍSO



UNIVERSIDAD
NACIONAL
DE COLOMBIA
SEDE BOGOTÁ



UNIVERSIDAD
CENTRAL



9 Higher Education Institutions | 1 Association



Fev. 2025 - Fev. 2028



799.887 Euros



Tech
Tra
PlastiCE



TechTraPlastiCE's Main Goal

🎯 To reinforce the **applied research and technological transfer capacities** of the universities improving their services portfolios to strengthen the plastic industry's contribution to the green transition In Colombia, Chile and Argentina.



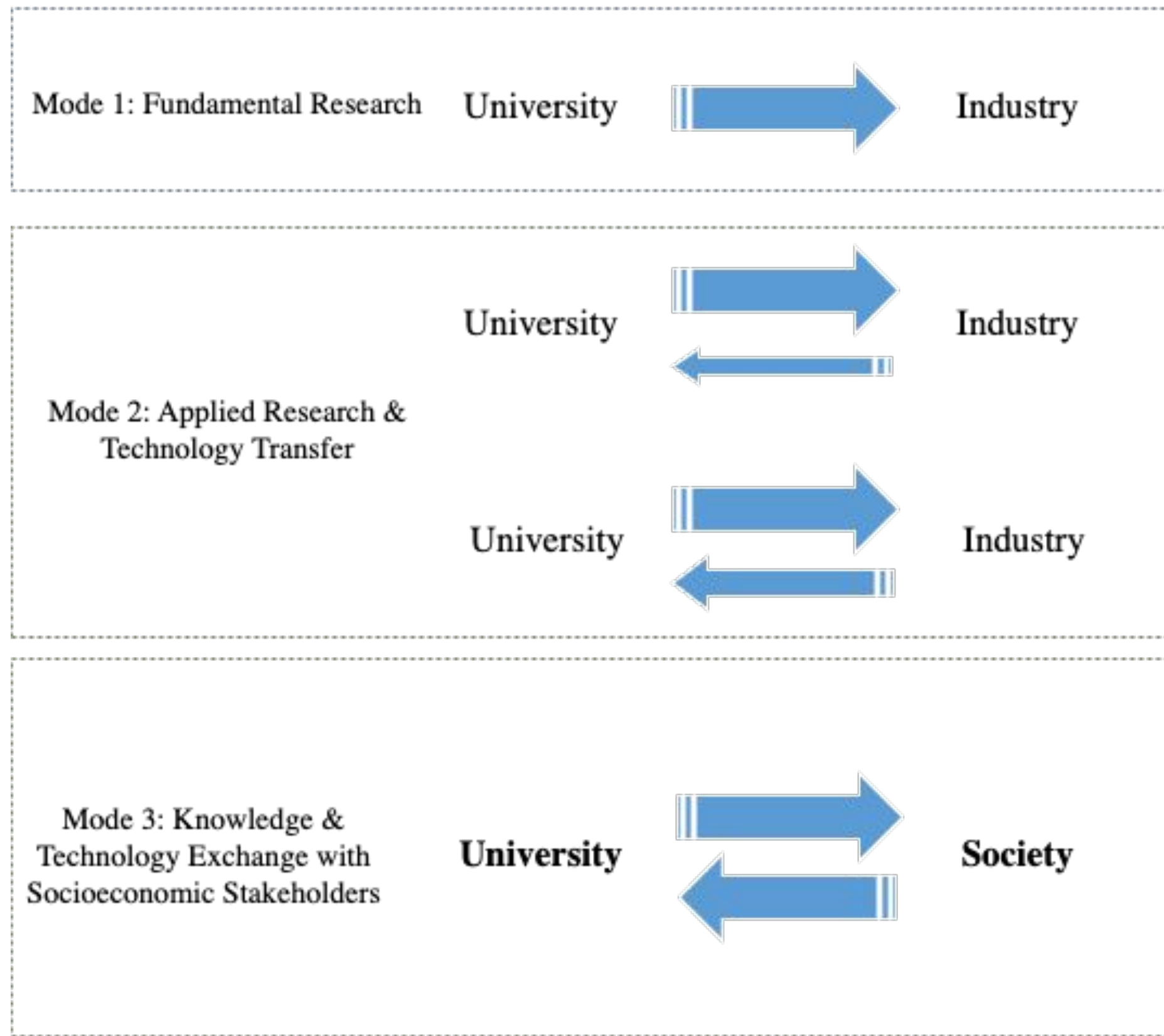
TechTraPlastiCE's Ambition



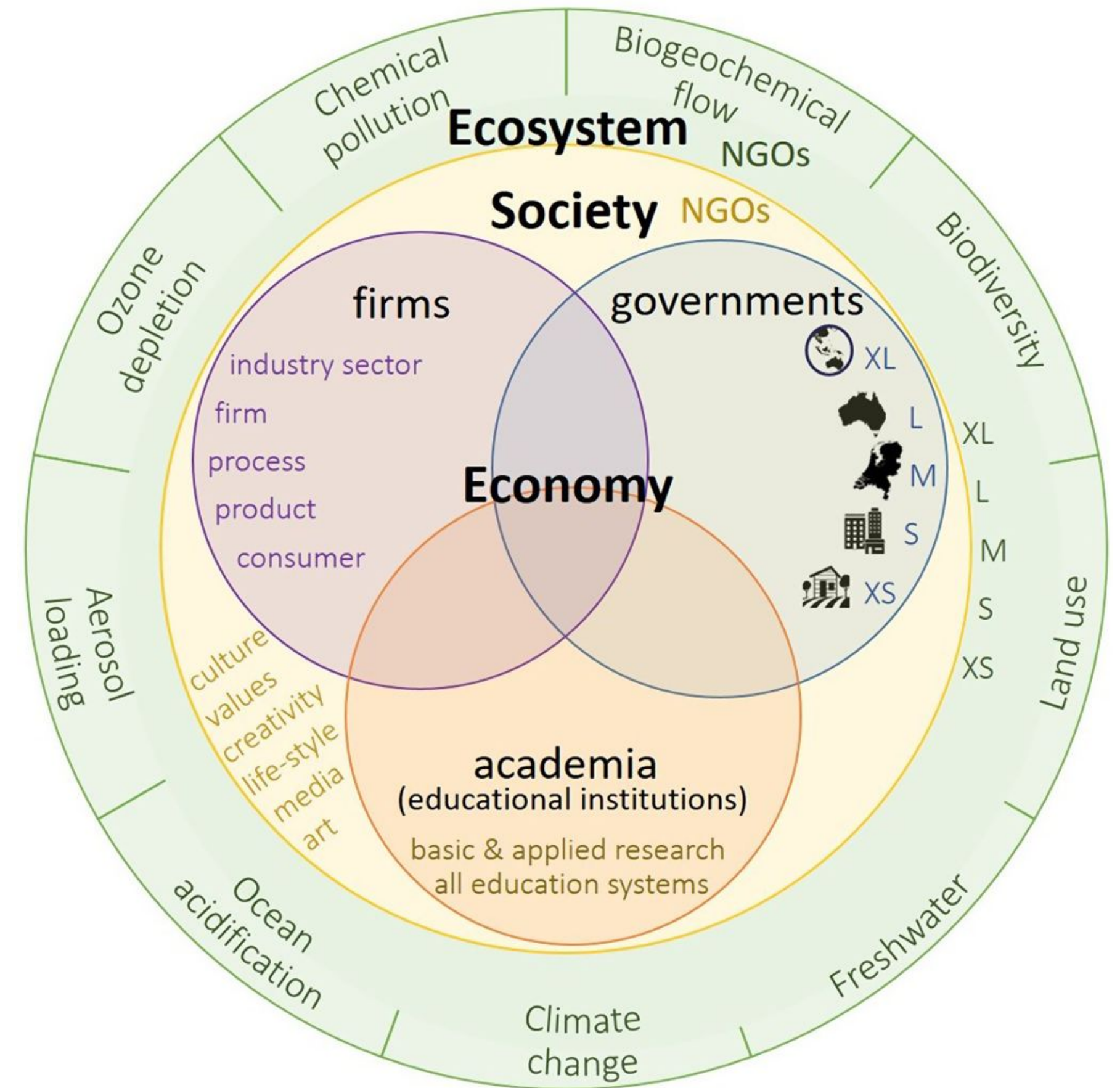
🎯 To foster the **collaboration and partnerships of the University with socio-economic stakeholders** to *incubate*, establish and develop circular initiatives.

We seek innovative strategies of *University-Enterprise collaboration* by carrying out joint recycling interventions integrating actors in the plastics value chain in Colombia, Argentina, and Chile

Modes of knowledge production & transfer



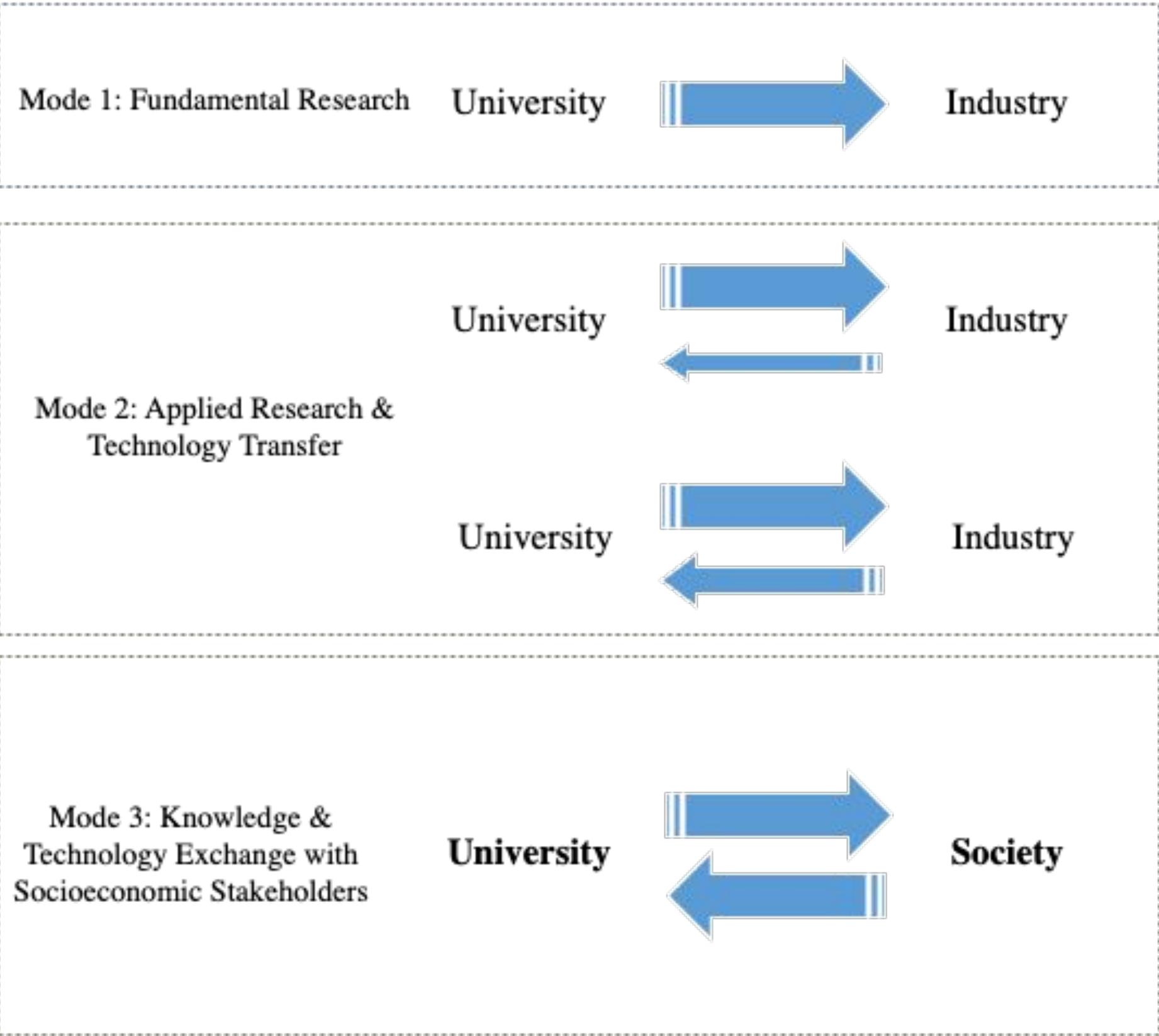
(Gibbons, 1994; Leonard-Barton, 1996)



van Bueren, B.J.A., Argus, K., Iyer-Raniga, U., Leenders, M.A.A.M., 2023. ***The circular economy operating and stakeholder model “eco-5HM” to avoid circular fallacies that prevent sustainability.*** Journal of Cleaner Production 391, 136096.
<https://doi.org/10.1016/j.jclepro.2023.136096>

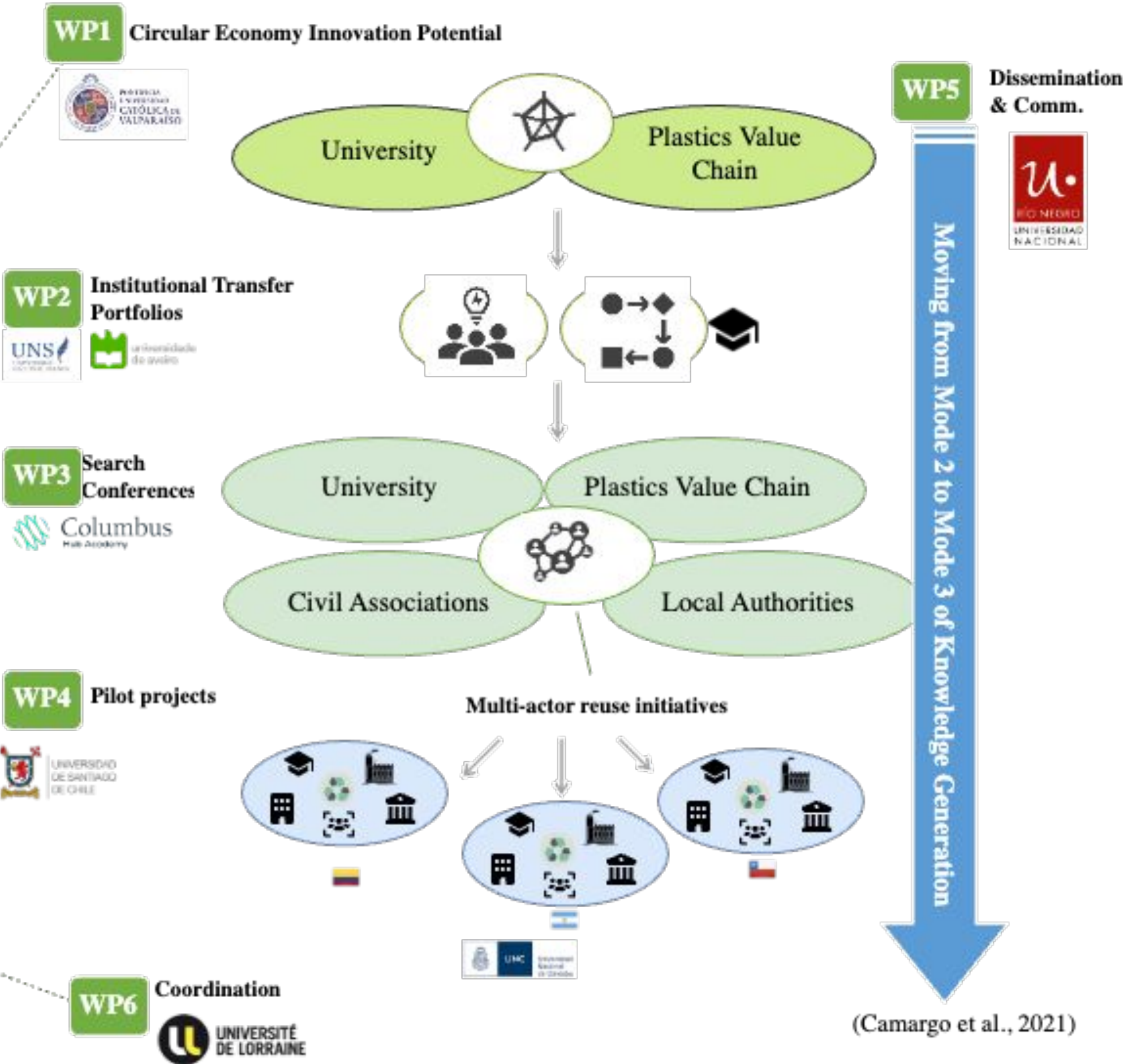
The Approach

Modes of knowledge production & transfer



(Gibbons, 1994; Leonard-Barton, 1996)

Progressive knowledge & technology transfer innovation capabilities to the local plastics value chain



(Camargo et al., 2021)



Analysis of Legislation framework of Circular Economy In Argentina, Colombia and Chile



Tech
Tra
PlastiCE



Co-funded by the
European Union

Legislation of CE

11 countries have launched government-led efforts to promote a circular economy for plastics (Rodríguez-Meza et al., 2026, p. 5)

: The roadmap for a Circular Chile by 2040

: National Circular Economy Strategy

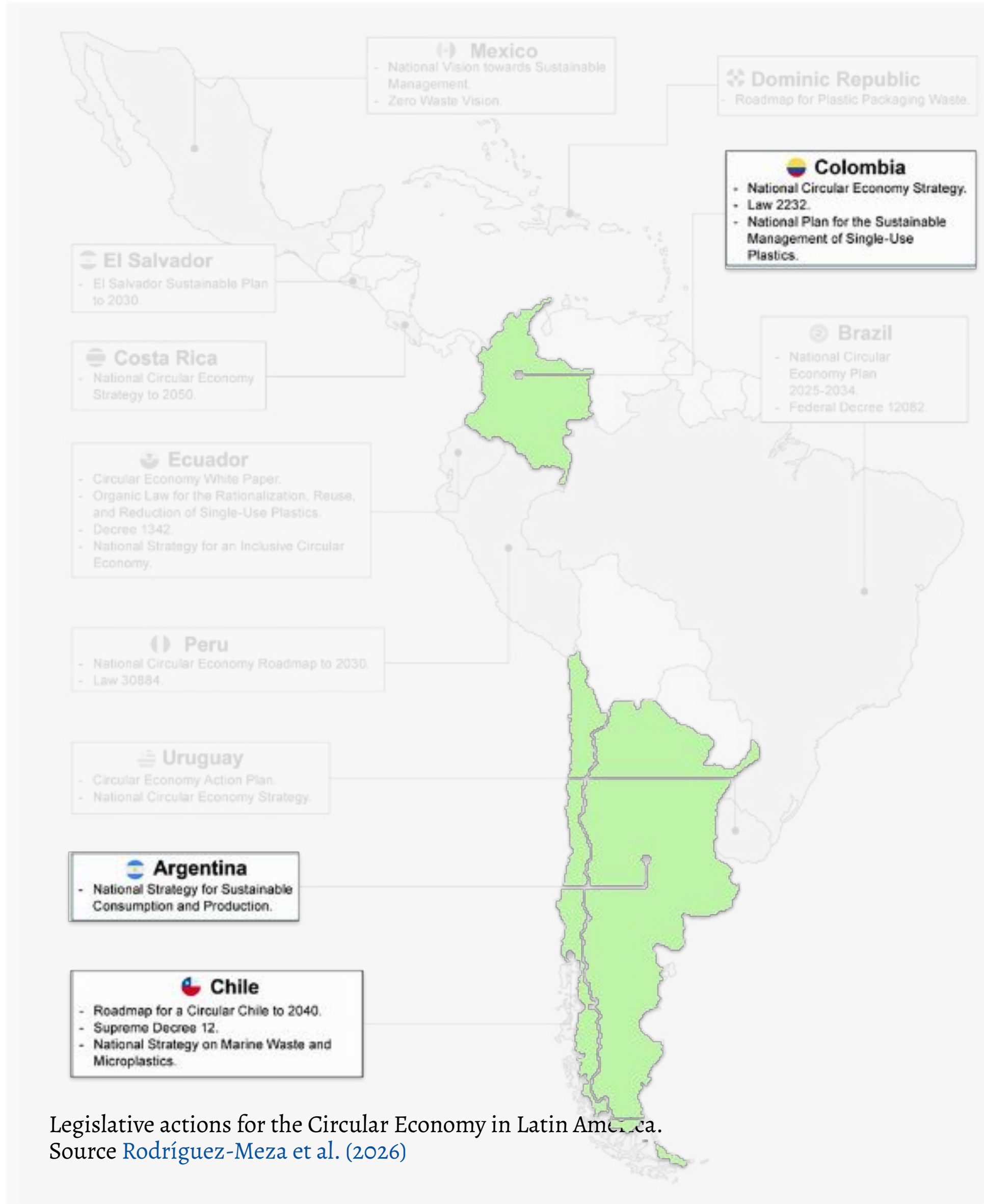
: National Strategy for Sustainable Consumption and Production



Legislation of CE

11 countries have launched government-led efforts to promote a circular economy for plastics (Rodríguez-Meza et al., 2026, p. 5)

Countries		Argentina	Chile	Colombia
Circular design	Recycled plastic and new (bio)polymers			
	Circular indicators and criteria			
	Chemicals harmful			
	Microplastics			
	Economic incentives			
Circular business models	Circular supplies			
	Product-as-a-service			
	Product life extension			
	Sharing platforms			
	Resource recovery			
The triple planetary crisis	Pollution			
	Biodiversity loss			
	Climate change			



Legislative actions for the Circular Economy in Latin America.
Source [Rodríguez-Meza et al. \(2026\)](#)

Legislation of CE

In conclusion,

Plastic problem is acknowledged..

However, the legal baseline is uneven..

Table 3.1: Regulatory And Policy Frameworks

Dimension	Argentina	Chile	Colombia
Plastic Regulatory Framework	It has general and sectoral environmental laws but lacks an integrated, plastic-specific regulatory framework at the national level. As a federal country, environmental regulation often falls under provincial jurisdiction, resulting in a patchwork of approaches and uneven implementation across regions.	Has a robust and specific legal framework regulating different stages of the plastic life cycle, including Extended Producer Responsibility (EPR) and single-use plastics.	It shows advances in specific laws on single-use plastics and EPR. However, the framework is still being developed and progressively regulated.
Circular Economy Public Policies	Some initiatives have been launched, but many are fragmented or discontinued. The lack of coordination across jurisdictions limits their long-term impact. Circular economy is referenced in laws and resolutions, but not yet integrated as a core policy framework; efforts vary by province.	Long-term strategies exist with measurable targets and clear guidelines. There is strong articulation among ministries and public actors. Circular economy is institutionalized with cross-cutting strategies like the 2040 Roadmap.	Active national circular economy strategy (2019) complemented by city-level efforts like Bogotá's circular policy (2023) and sectoral plans. Circular economy is integrated across government levels and sectors, with special emphasis on inclusion, formalization of recyclers, and value chain innovation.
Participation in International Agreements	Recognizes and signs international agreements but are not always translated into concrete or binding national actions.	Active participant in multilateral environmental agreements; uses these as leverage for national policy development.	Strong engagement in international environmental agreements, some of which form the basis for new local policies and frameworks.
Implementation and Coordination	Policy implementation is uneven, with local initiatives often disconnected from national agendas; limited fiscal incentives and regulatory alignment.	Strong legal base, but implementation gaps remain due to limited infrastructure in some regions and challenges in local enforcement.	Progress has been made in legislation and planning, but challenges persist in financing, local capacity building, and institutional strengthening.



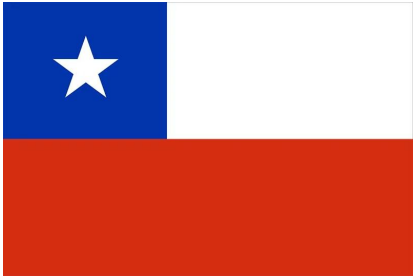
Engaging with the Plastic Value Chain in Argentina, Colombia and Chile

Engaging with Unions and Associations

- 1. Analysis of Legislation framework of Circular Economy.
- 2. Evaluation of the Innovation Capacity for Plastic Industrie at AR, CH, CL
 - About 80 Enterprises
- 3. University Portfolio



Acoplásticos



(a) Argentina



(b) Chile



(c) Colombia



techtraplastice.eu

@techtraplastice



Maturity Grids on Innovation

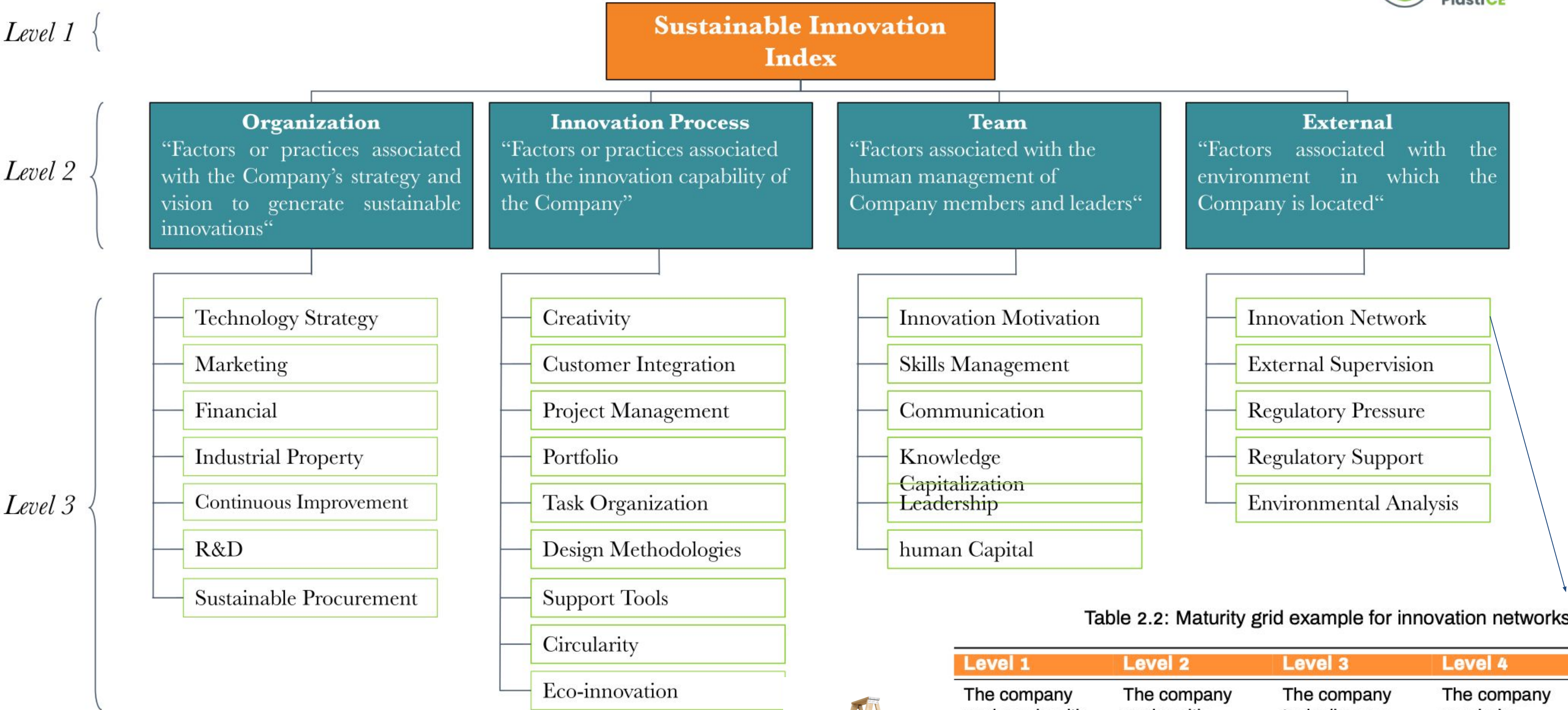


Table 2.2: Maturity grid example for innovation networks –IN-.

Level 1	Level 2	Level 3	Level 4	Level 5
The company works only with internal resources	The company works with customers or suppliers.	The company typically uses external resources privileging two of the three impacts	The company regularly engages in collaborative projects that are selected according to its sustainable innovation strategy.	The company follows an openness strategy with leading actors in sustainability.

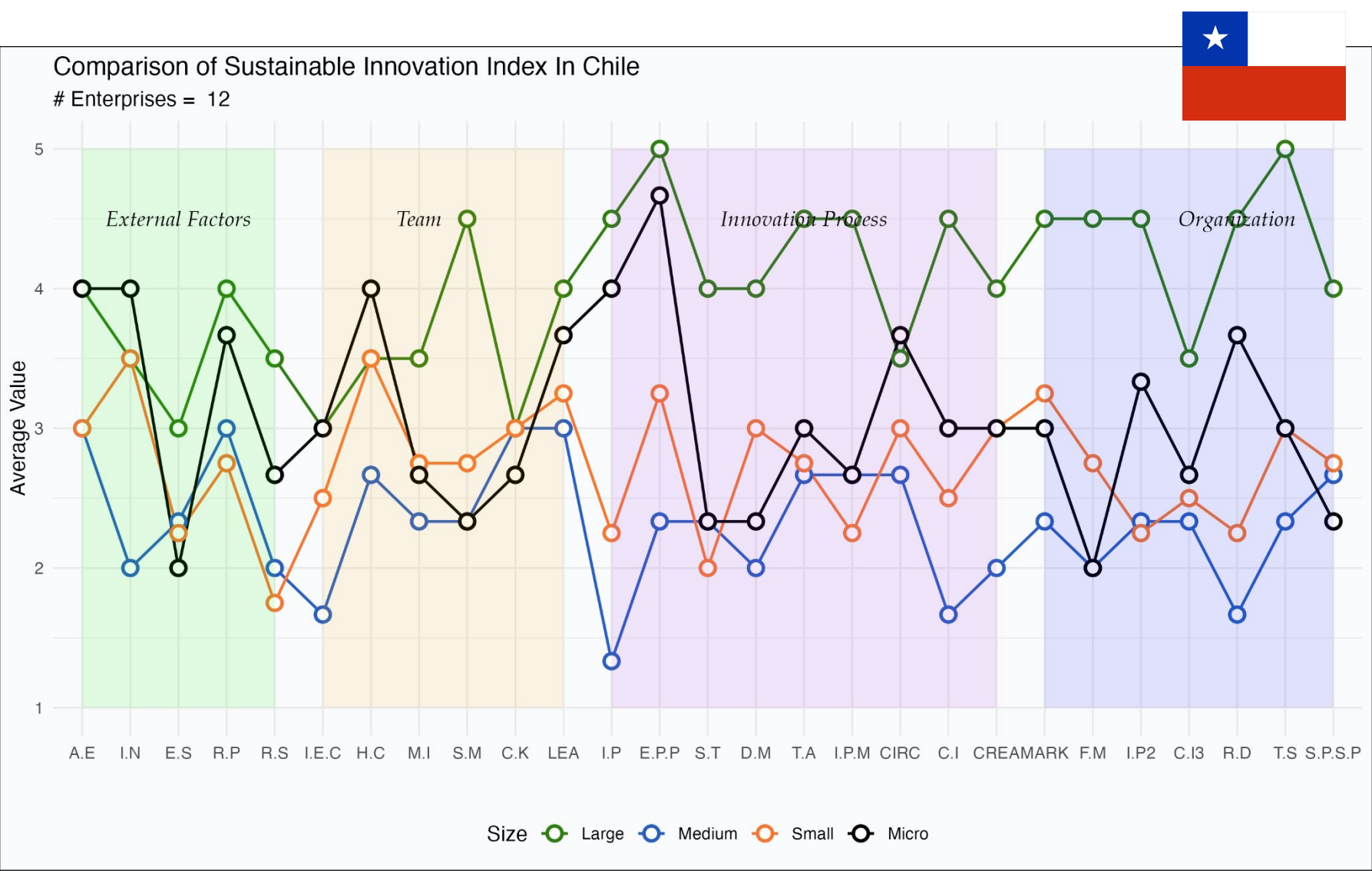
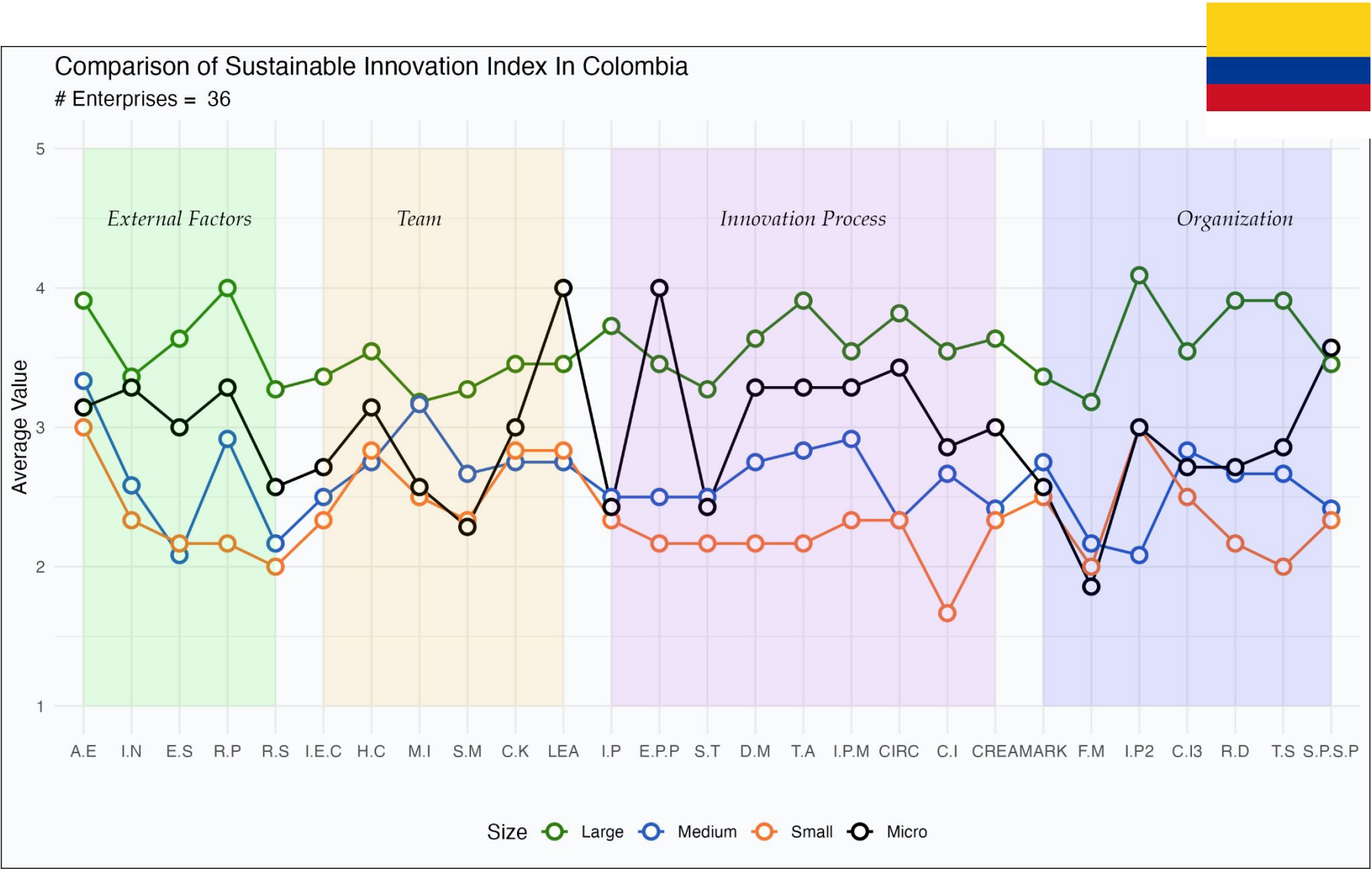
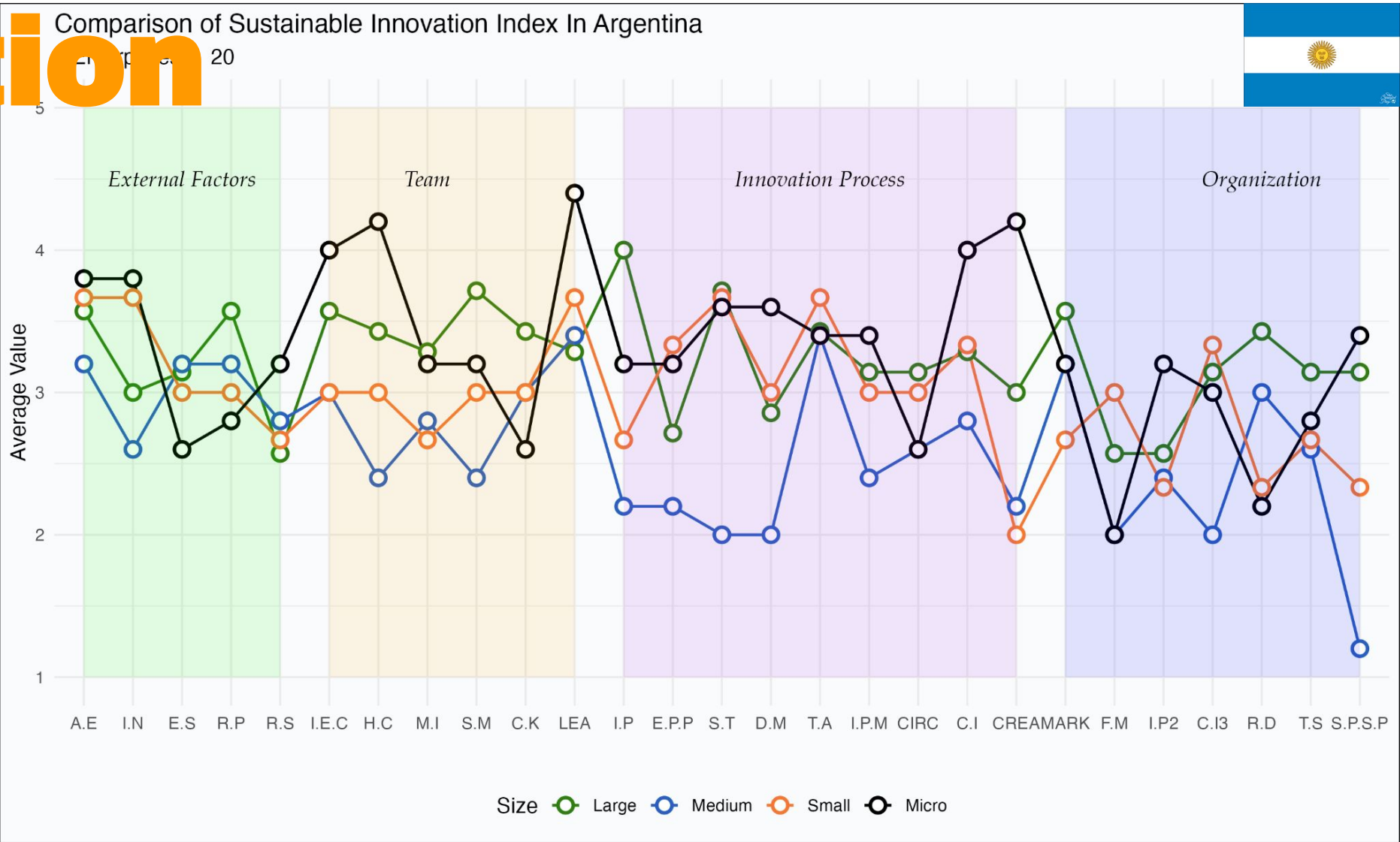
Astudillo-Ortiz, Matías, et al. "Analysis of Sustainable Innovation Management in the Chilean Plastics Industry: Towards a Taxonomic Approach Based on Internal and Contextual Factors." International Association for the Management of Technology Conference. Cham: Springer Nature Switzerland, 2025.



Maturity Grids on Innovation

Table 3.1: Distribution of participating companies by size

Country	Size	# Enterprises	% per country
Argentina	Large	7	35.00
	Medium	5	25.00
	Small	3	15.00
	Micro	5	25.00
Chile	Large	2	16.67
	Medium	3	25.00
	Small	4	33.33
	Micro	3	25.00
Colombia	Large	11	30.56
	Medium	12	33.33
	Small	6	16.67
	Micro	7	19.44





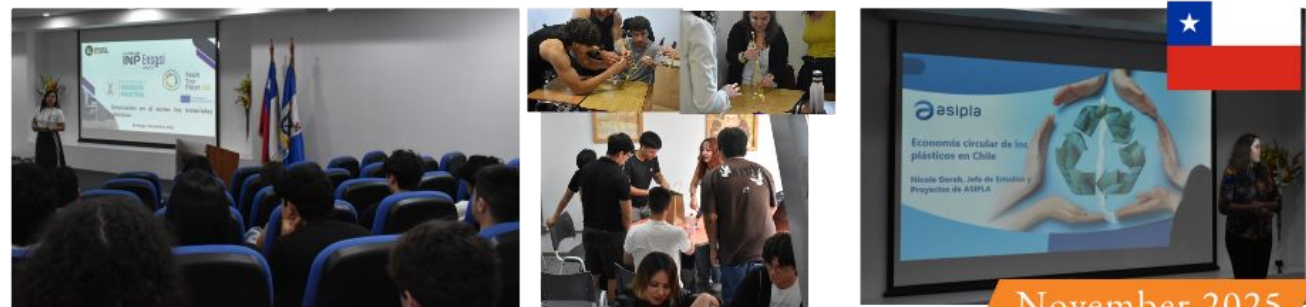
Multi-actor collaborations for just and safe plastic's circular economy

Academia - Industry: Co-creation as strategy

Academia as explorer of the fuzzy front end of Innovation



August 2025



November 2025

HEI



External Partner



August 2026

Implementation of Innovation Challenges with Students

Academia as a Facilitator of Multi-Sector Collaboration



SEARCH CONFERENCE

24 de julio de 2026
Universidad Central (UC)
Bogotá, Colombia



techtraplastice.eu

@techtraplastice



Implementation of Search Conference methodology as strategic planning tool

Innovation Challenges Academia - Industry

ORION

Osez la recherche !

FRANCE

2030

LORRAINE

INP

Ensgsi

NANCY

UNIVERSITÉ DE LORRAINE

COLLEGIUM LORRAINE

MANAGEMENT INNOVATION

48h

pour faire
vivre des idées

édition 2025

Développez votre créativité

Du 26 au 28 novembre 2025



Co-creation Academia - Industry

User-Centered Co-Creation:
Stakeholders are active partners in
design rather than just final testers.



UNIVERSIDAD
NACIONAL
DE COLOMBIA



UNIVERSIDAD
CENTRAL

Acoplásticos



Feedback of the Search Conference Bogota

Search Conference Methodology

- ❑ Opening Presentations
- ❑ EPR Review: Chile (PUCV) & Colombia (Acoplásticos)
- ❑ Panel Discussion: Key topics of interest
- ❑ Working Groups: Co-creation of initiatives
- ❑ Group Conclusions
- ❑ UC Rector's closing Remarks

81 Participants Representing:

- ❑ 5 Government and Public Institutions
- ❑ 7 Plastics Manufacturers and Processors
- ❑ 2 Industry Associations (CCB and Acoplásticos)
- ❑ 3 Waste Picker Organizations
- ❑ 3 External Universities
- ❑ 1 Plastic-Using Company / Brand
- ❑ TechTraPlastiCE team

Group 1: Business Models (Pyrcom, ICIPC, Gescol, Kuiica, ANLA, Biocírculo, CCB)

Key Collaborative Initiatives:

- Connect R&D&I supply and demand.
- Develop a platform for valorized plastic products.
- Facilitate strategic project development.

Group 2: Knowledge and Technology Generation and Transfer (Acoplásticos, UC DIT, Uniandes, ICIPC, AsoreWay, Ecoplaneta, Kuiica)

Key Collaborative Initiatives:

- Establish an Academia–Industry Coordination Committee (Supported by Acoplasticos).
- Create an information system to connect capabilities and research.
- Promote industry-oriented university R&D and eco-design.

Group 3: Infrastructure Development (Bio-Círculo, SOPLASCOL, ANLA)

Key Collaborative Initiatives:

- Develop coordinated consumer awareness programmes.
- Strengthen collection, separation and traceability infrastructure.

Group 4: Education and Training (UAESP, INVIMA, Gobernación Cundinamarca, Pinturas Bler, UC, U Nacional, AsoreWay)

Key Collaborative Initiatives:

- Develop multi-stakeholder training pilots.
- Promote industrial symbiosis in Cundinamarca.
- Develop policy papers to support public decision-making.

Major difficulties in the path!

1. Trust-building process takes time.
2. Difficulties at the Internal integration (silo effect)
3. Role of teacher as co-production of knowledge.
4. Role of Academia as a potential coordination hub connecting industry, government, waste picker organizations, and plastics manufacturers and processors.



**Tech
Tra
PlastiCE**

**Strengthening University tech
transfer capabilities to support
circular economy value chains
for plastics in Latin America**

Thank you for you attention!

Assistant Prof. Fabio A. Cruz Sanchez
General Coordinator
Sept 15/2026



Interreg

Grande Région | Großregion



Cofinancé par
l'Union Européenne
Kofinanziert von
der Europäischen Union



CIRKLA
UniGR | Centre for Circular Economy
of Metals and Materials



Co-funded by
the European Union

Feedback of the Search Conference Bogota

Strategic Convergences Among Groups:

Academia–Industry Collaboration: Connect industry needs with university R&D capabilities.

Training & Eco-design: Develop specialized talent and support material circularity.

Traceability Across the Value Chain: Strengthen information systems and end-to-end data management.

Conclusions

The Search Conference highlighted the importance of thorough **preparation** and **sustained stakeholder engagement** to ensure the participation of relevant actors and enrich collective proposals.

Main challenges include collaboration, **knowledge transfer**, traceability, **infrastructure**, financing, and training. The proposed initiatives position academia as a potential coordination hub connecting industry, government, waste picker organizations, and plastics manufacturers and processors.

Connect R&D&I supply and demand, develop information and **traceability systems**, strengthen circular economy education, and create permanent spaces for **multi-stakeholder projects** and pilot implementation.

UC Pilot Project Outline

- ❑ Internal Review of the Action Plan – Faculty of Engineering and Basic Sciences (MOOC Course - NexusPlast - UC Sustainable Environmental Management System)
- ❑ Follow up on the identified strategies with Universidad Nacional de Colombia and Acoplásticos.
- ❑ Participate in the National Roundtable for Sustainable Plastics Management in Colombia as an invited participant.
- ❑ Explore the development of an internal training guide for public institutions on Law 2232 with Ministerio de Ambiente.

Relationship effect

- ❑ Invitation Clúster de Empaques ICIPC- REPs en LATAM
- ❑ U Antioquia-ANLA collaboration for a reserach project to evaluate impact of single use plastic law.



**Tech
Tra
PlastiCE**

**From sustainability policy to
strategic innovation: analyzing
circular maturity and business
transformation in Colombia's
plastic value chain**

ECHO 2026 – Sustainability and the Circular Future with UniGR-CIRKLA

Aquileo Piracoa Hernández, Paulo César Narváez Rincón, Fabio Cruz, Mauricio Camargo Pardo,
Álvaro Orjuela Londoño



Interreg

Grande Région | Großregion



Cofinancé par
l'Union Européenne
Kofinanziert von
der Europäischen Union



CIRKLA
UniGR | Centre for Circular Economy
of Metals and Materials



Co-funded by
the European Union

Plastic Problem: View from Colombia

In Colombia, while a robust regulatory landscape has evolved, actual value chains remain highly linear.

Structural Barriers:

1. Underfunded technology transfer offices
2. Misalignment between academic outputs and industry needs
3. Industry needs hinder effective collaboration.



Goal & Approach

Identification of the barriers and lock-ins to Technology Transfer within Colombia's plastics value chain.

A significant gap persists between knowledge generation and its effective implementation in the productive sector

The study employs a mixed qualitative-quantitative approach that includes an assessment of the productive sector through surveys of 92 companies and semi-structured interviews with key stakeholders



UNIVERSIDAD
NACIONAL
DE COLOMBIA



UNIVERSIDAD
CENTRAL

Acoplásticos

Methodology

Phase I Documentary Analysis

**Legal framework
of CE within
Colombian
context**



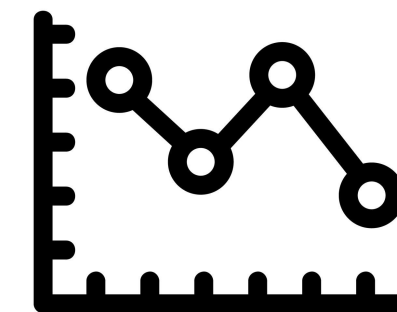
Phase II Product sector diagnosis (Interviews and surveys)

**Exploratory
survey to
companies and
semi-structured
interviews**



Phase III Evaluation of Circular Maturity (survey)

**Innovation index
determined
based on
companies
self-diagnosis**



Phase IV Validation and strategy design proposals

**48-h co-creation
workshop with
students and
industry experts**



Key finding: The sustainability paradox

There is a significant gap between the formalization of corporate sustainability policies and the existence of structured innovation processes.

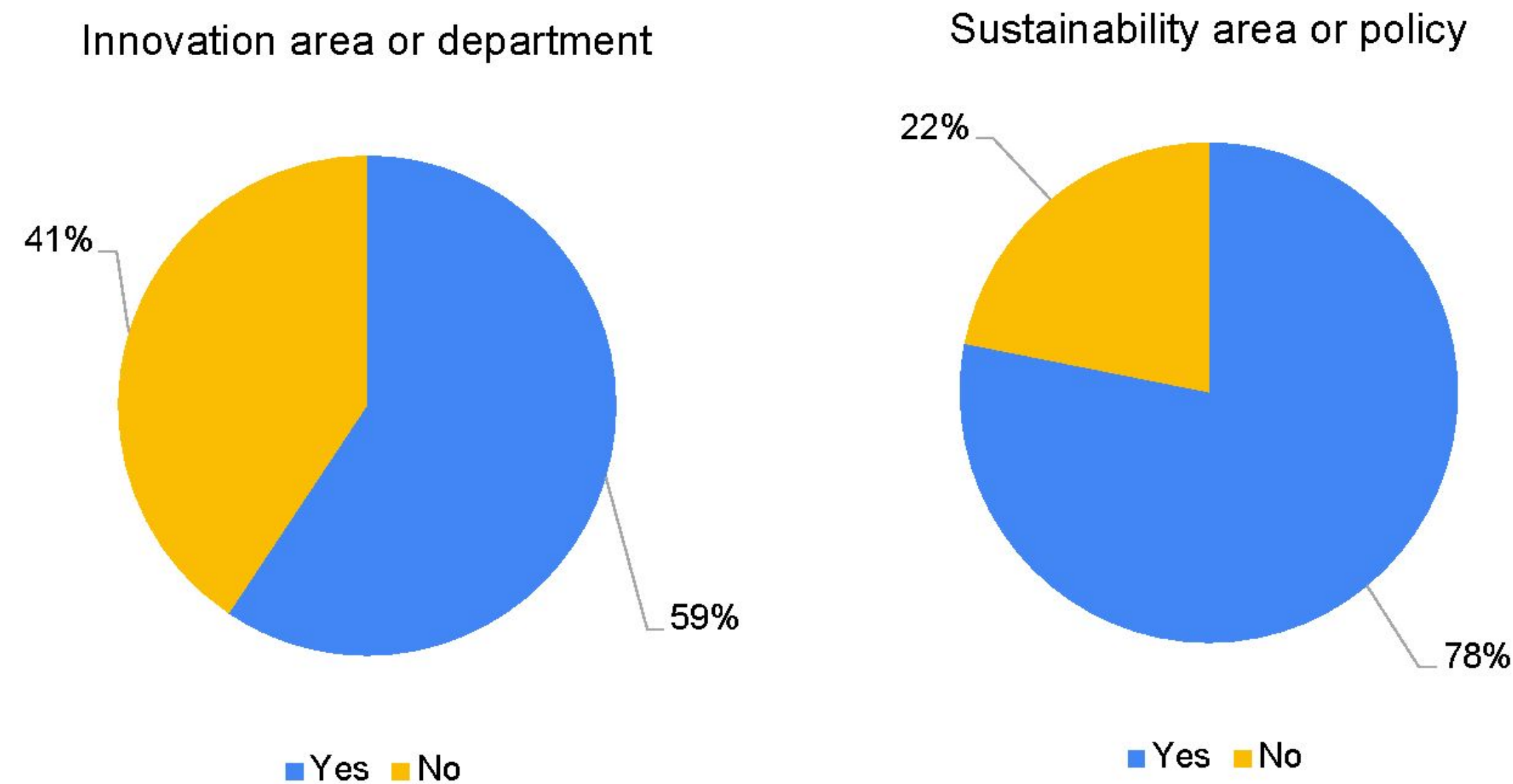
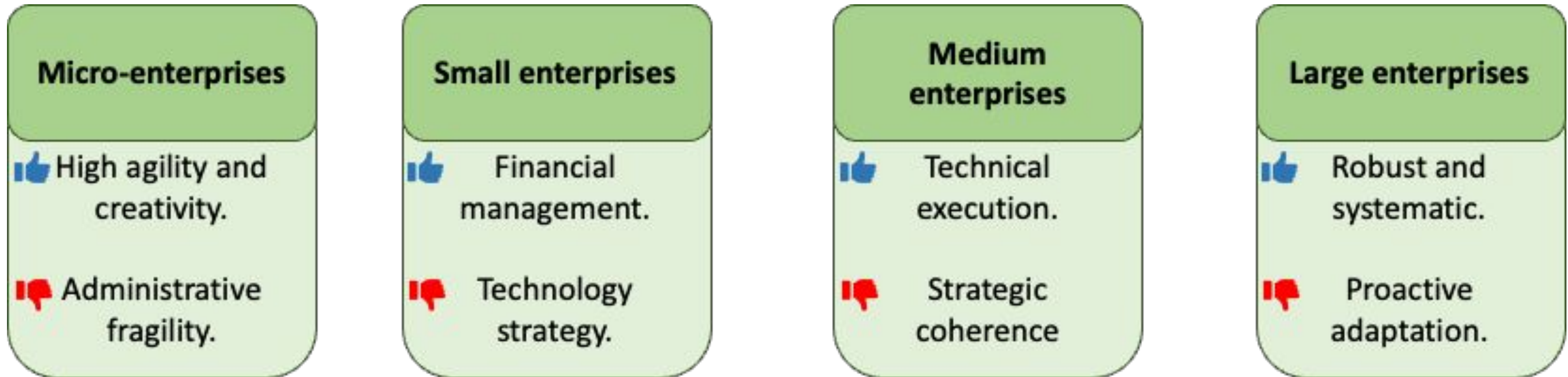


Figure 1. Companies that reported having an innovation area or department, and a sustainability area or policy.

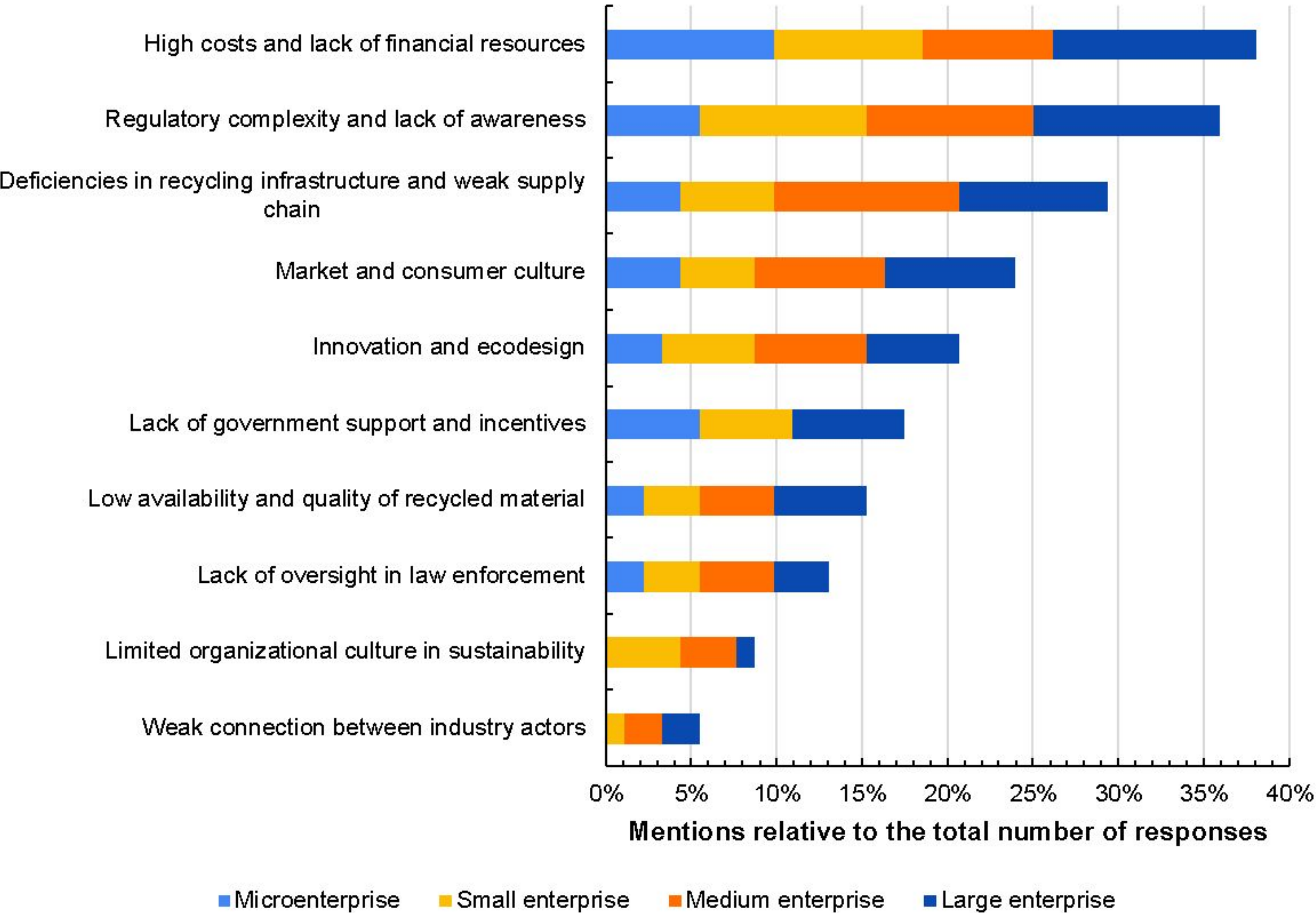
Key finding: Asymmetries in circular maturity

Different maturity profiles towards circular economy practices by enterprise size.



Key finding: Systemic Barriers

Figure 2. Most frequently reported barriers to compliance with plastics and circular economy regulations, and their relevance according to company size.



Key finding: Academia - Industry Alignment

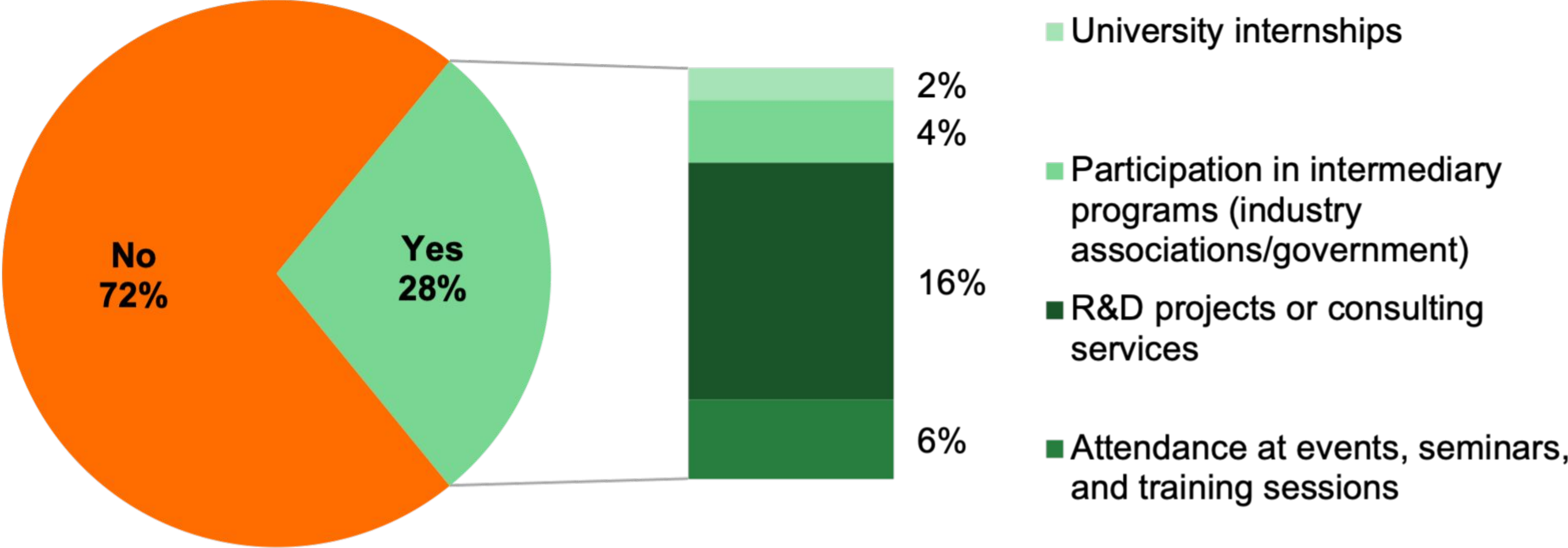
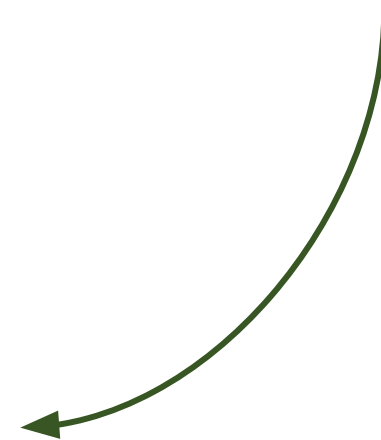


Figure 3. Companies reporting relationships with academia and types of collaboration.

Opportunities : Academia - Industry

Emerging opportunities exist in product ecodesign and digital value chain integration to transform plastic from an environmental liability into a high-value resource.

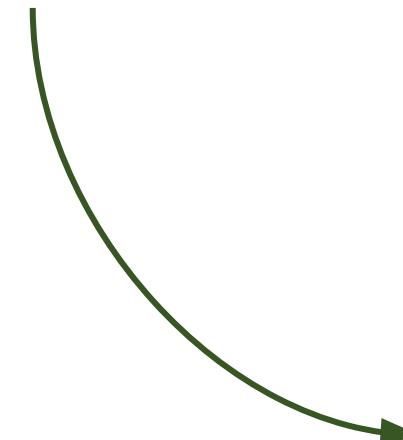
***Practical
ecodesign***



***Industrial
symbiosis***



***Value chain
formalization***



Conclusions

- The plastics sector in Colombia is dominated by packaging and intermediate transformation activities.
- The main barriers to implementing circular economy practices include high costs of recycled materials, insufficient and informal recycling infrastructure.
- Technology-transfer barriers originate from minimal university–industry interaction.
- To enhance alignment between academia and industry is required structured collaboration platforms, targeted capacity-building programs, and projects focused on applied research and ecodesign.



**Tech
Tra
PlastiCE**

**From sustainability policy to
strategic innovation: analyzing
circular maturity and business
transformation in Colombia's
plastic value chain**

Thank you for your attention!

Aquileo Piracoa Hernández, Paulo César Narváez Rincón, Fabio Cruz, Mauricio Camargo Pardo,
Álvaro Orjuela Londoño



Interreg

Grande Région | Großregion



Cofinancé par
l'Union Européenne
Kofinanziert von
der Europäischen Union



CIRKLA
UniGR | Centre for Circular Economy
of Metals and Materials



Co-funded by
the European Union